TEXORA · LECTURE SHEET

Analytic Functions

Functions of a complex variable, differentiability, the Cauchy–Riemann equations as the necessary and sufficient test, harmonic functions and their conjugates, orthogonal families, and Milne–Thomson construction.

MATH-1201 · MATHEMATICS-II · TOPIC 4 · LEVEL 1 TERM 2

1. Functions of a complex variable

With z = x + iy, a function w = f(z) splits into two real functions of two real variables:

$$w = u(x, y) + i\, v(x, y)$$

Examples: f(z) = z² gives u = x² − y², v = 2xy; f(z) = e^z gives u = e^x cos y, v = e^x sin y. The **derivative** is defined as in real calculus, f′(z) = lim_{Δz→0} Δw/Δz — but Δz can now approach along *any* path in the plane, and the limit must be the same along all of them. That single demand is far stronger than real differentiability; it is what makes complex functions so rigid and so powerful. A function differentiable at every point of a region is **analytic** there (also called regular or holomorphic).

2. The Cauchy–Riemann equations

Approach along the real axis (Δz = Δx): f′ = u_x + i v_x. Approach along the imaginary axis (Δz = iΔy): f′ = v_y − i u_y. Equality of the two limits forces

$$\partial u/\partial x = \partial v/\partial y \quad \text{and} \quad \partial u/\partial y = -\partial v/\partial x \quad \text{(the Cauchy–Riemann equations)}$$

Necessary: every analytic function satisfies CR at every point of its region. **Sufficient:** if u, v have continuous first partial derivatives in a region and CR holds throughout it, then f is analytic there (the syllabus asks for the statements, proofs excluded). Differentiability at a single point is *not* analyticity: analyticity requires CR throughout a neighbourhood.

WORKED TESTS

f(z) = z²: u = x² − y², v = 2xy. u_x = 2x = v_y ✓; u_y = −2y = −v_x ✓ (v_x = 2y). CR everywhere → analytic everywhere, f′(z) = u_x + iv_x = 2x + 2iy = 2z ✓.

f(z) = z̄: u = x, v = −y. u_x = 1 but v_y = −1 → CR fails everywhere → **nowhere analytic** (conjugation mirrors the plane; it cannot be a complex derivative).

f(z) = |z|²: u = x² + y², v = 0. CR demands 2x = 0 and 2y = 0: holds only at the origin.

Differentiable at 0 ($f'(0) = 0$), analytic **nowhere** — the standard warning that CR at a point ≠ analytic.

3. Harmonic functions and conjugates

If f is analytic, u and v each satisfy **Laplace's equation**:

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0$$

(differentiate the CR equations and subtract). Functions satisfying Laplace's equation are **harmonic**. Given a harmonic u, a harmonic v satisfying CR with it is its **harmonic conjugate**, found by integrating CR:

WORKED CONSTRUCTION

$u = x^2 - y^2$ (harmonic: $u_{xx} + u_{yy} = 2 - 2 = 0$ ✓). Find v: $v_y = u_x = 2x \rightarrow v = 2xy + g(x)$.
Then $v_x = 2y + g'(x)$ must equal $-u_y = 2y$, so $g' = 0$, $g = C$.
$v = 2xy + C$, and $f(z) = (x^2 - y^2) + i(2xy) + iC = \mathbf{z^2 + iC}$ — the conjugate rebuilds the function the parts came from.

Orthogonal property: the families of curves $u = c_1$ and $v = c_2$ cut each other at right angles wherever $f'(z) \neq 0$ (their gradients obey $\nabla u \cdot \nabla v = u_x v_x + u_y v_y = 0$ by CR). For z^2 these are the two families of rectangular hyperbolas $x^2 - y^2 = c_1$ and $2xy = c_2$ — visibly perpendicular.

4. Milne–Thomson method

To construct f from u alone (or v alone) without integrating: write $u_x = \varphi_1(x, y)$, $u_y = \varphi_2(x, y)$. Then

$$f'(z) = \varphi_1(z, 0) - i\,\varphi_2(z, 0), \quad \text{integrate for } f(z)$$

WORKED EXAMPLE

Given $u = x^2 - y^2 + x$. $u_x = 2x + 1 \rightarrow \varphi_1(z, 0) = 2z + 1$; $u_y = -2y \rightarrow \varphi_2(z, 0) = 0$.
$f'(z) = 2z + 1 \rightarrow f(z) = \mathbf{z^2 + z + C}$. Check: $\mathrm{Re}(z^2 + z) = x^2 - y^2 + x$ ✓.

5. Physical applications

Any 2-D field governed by Laplace's equation inherits this machinery through the **complex potential** $\Omega(z) = \varphi + i\psi$: for steady, incompressible, irrotational fluid flow, φ is the velocity potential and ψ the stream function — CR is exactly the statement that flow is both irrotational and incompressible, and the orthogonal families are the equipotentials and streamlines. The same pair serves electrostatics (φ =

potential, ψ = field's flux function) and steady heat conduction (φ = temperature). This is why harmonic functions are called the mathematics of equilibrium fields.

6. Key points

- Analytic = differentiable throughout a region; the derivative's path-independence is the source of the rigidity.
- CR: u_x = v_y, u_y = −v_x. Necessary always; sufficient when the partials are continuous and CR holds in a region.
- u, v of an analytic function are harmonic conjugates of each other; each satisfies Laplace's equation.
- The families u = c, v = c′ are orthogonal (where f′ ≠ 0).
- Construction routes: integrate CR directly, or Milne–Thomson ($f'(z) = \varphi_1(z, 0) - i\varphi_2(z, 0)$).
- f′(z) can be read from any CR pair: f′ = u_x + iv_x = v_y − iu_y.

7. Practice

Answer in your own words; brief model answers follow.

1. Determine whether $f(z) = x^2 + iy^2$ is analytic anywhere.
2. Find the harmonic conjugate of $u = x^3 - 3xy^2$, and the analytic function f(z) it belongs to.
3. Find the harmonic conjugate of u = e^x sin y, and express f(z) compactly.
4. State the Cauchy–Riemann equations in polar form and use them to check that log z (principal branch) is analytic where defined.
5. Prove: if f and its conjugate $\bar{f}$ are both analytic in a region, then f is constant there.
6. Using Milne–Thomson, construct the analytic function whose imaginary part is v = 2xy + x.

Answer key

1. $u = x^2$, $v = y^2$: u_x = 2x, v_y = 2y — equal only on the line y = x; u_y = 0 = −v_x ✓. CR holds only along a line, never in a region, so f is analytic **nowhere** (differentiability on a line does not make analyticity).
2. $v_y = u_x = 3x^2 - 3y^2 \rightarrow v = 3x^2y - y^3 + g(x)$; $v_x = 6xy + g'(x) = -u_y = 6xy \rightarrow g = C$. $v = 3x^2y - y^3 + C$, and $f(z) = z^3 + iC$ (since Re $z^3 = x^3 - 3xy^2$ ✓).
3. v_y = u_x = e^x sin y → integrating in y, v = −e^x cos y + g(x); check v_x = −e^x cos y = −u_y (since u_y = e^x cos y) ✓ with g constant. v = −e^x cos y + C. Then u + iv = e^x sin y − ie^x cos y = −ie^x(cos y + i sin y), so f(z) = −ie^z + iC.
4. Polar CR: $\partial u/\partial r = (1/r)\,\partial v/\partial\theta$ and $\partial v/\partial r = -(1/r)\,\partial u/\partial\theta$. For log z: u = ln r, v = θ. $\partial u/\partial r = 1/r = (1/r)(1) = (1/r)\partial v/\partial\theta$ ✓; $\partial v/\partial r = 0 = -(1/r)(0)$ ✓. CR holds wherever log z is defined (r > 0, θ on the

chosen branch) → analytic there, derivative 1/z.

5. If f is analytic, CR: $u_x = v_y$, $u_y = -v_x$. If $\bar{f} = u - iv$ is analytic, its CR pair reads $u_x = -v_y$ and $u_y = v_x$. Both together force $u_x = v_y = 0$ and $u_y = v_x = 0$ throughout the region: u and v are constants, so f is constant. (Moral: conjugation destroys analyticity unless there is nothing to destroy.)
6. Milne–Thomson from v: with $\psi_1 = v_x = 2y + 1$, $\psi_2 = v_y = 2x$, the formula from the imaginary side is $f'(z) = \psi_2(z, 0) + i\psi_1(z, 0) = 2z + i(1) = 2z + i$. Integrate: $f(z) = z^2 + iz + C$. Check: $\mathrm{Im}(z^2 + iz) = 2xy + x$ ✓ (taking C real).
